

Research papers

Hydraulic conductivity estimation by NMR data in unconsolidated geological materials: insights from SDR model calibrations

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ABSTRACT

Accurate hydraulic conductivity (K) estimation is essential for effective groundwater resource management, yet conventional methods struggle with spatial heterogeneity and scale disparities. This study evaluated nuclear magnetic resonance (NMR)-based K prediction using the Schlumberger-Doll Research (SDR) model across diverse unconsolidated environments. We analyzed 92 laboratory samples from nine sites and 198 borehole screens at eight sites in Denmark and Germany, encompassing sediments (sand, clay, till) and organic deposits (peat, gyttja). Bootstrapping and statistical analysis revealed that general calibration of the SDR model was successful for inorganic sediments but failed when organic geological units were present in the dataset. No correlation was found between the mean logarithm of the NMR transverse relaxation time distribution (T_{2ML}) and K in peat soils, making reliable K estimation impossible in these materials. However, gyttja – despite being an organic soil – allowed for reliable K estimation using geology-specific calibration, although with different SDR constant coefficients than those for inorganic sediments. Generally, geology- or site-specific calibrations significantly improved K accuracy for inorganic geological units when sufficient data points were available for the curve fitting, and T_{2ML} and K values spanned a sufficiently broad range (e.g., at least 1 to 1.5 orders of magnitude). We demonstrated that using a simplified SDR model – excluding porosity ($K = bT_{2ML}^n$) – for unconsolidated inorganic geological materials reduces uncertainty in the free-fitting coefficient parameters (b and n). Moreover, it avoids significant errors arising from magnetic susceptibility-induced porosity underestimation, which reached up to 70 % in our samples.

1. Introduction

Hydraulic conductivity (K) is one of the key hydrogeological properties of subsurface materials and defines how easily liquids flow through porous media (Shackelford, 2013). Quantitative measures of K are a prerequisite for describing groundwater flow rates, solute fluxes, as well as the spatial and temporal distribution of hydraulic heads (Woessner and Poeter, 2020). Hydraulic conductivity plays a critical

role in a wide range of applications, from geotechnical engineering, where it influences soil stability and drainage behavior (Fredlund and Rahardjo, 1993; Zhang and Chen, 2005), to environmental studies, where it determines contaminant transport and effectiveness of remediation strategies (Gelhar et al., 1992). In groundwater resource management, quantifying K is essential for assessing aquifer productivity, designing wells, and predicting the impact of water extraction on surrounding hydrogeological systems (Anderson et al., 2015; Zhou and Li,

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2011). Given its importance, accurately estimating K is a key objective for hydrogeological characterization and modeling.

One challenge faced by hydrogeologists is that estimating K at the field scale is necessary for predicting groundwater flow. However, K is typically determined from laboratory data or empirical predictions based on small-volume samples that do not represent the field scale (Woessner and Poeter, 2020) and may not reflect in situ pore conditions in the field. Even when in situ K data are collected using field measures such as aquifer pumping tests or slug tests, the data is often sparse with a coarser than desired resolution (e.g., a single K estimate per borehole or per screened interval). An alternative tool in this context is Nuclear Magnetic Resonance (NMR), specifically Borehole NMR (BNMR), which bridges the gap between the two previously mentioned K measurement approaches by providing high-resolution in situ K estimates (Walsh et al., 2013). The NMR tool can be used in shallow and relatively narrow-diameter boreholes and piezometers and can also be deployed via the direct-push approach without the need to perform permanent borehole/piezometer installations (Morozov et al., 2024). Several field-based studies of NMR-based K estimation in unconsolidated geological materials have been conducted (Crow et al., 2021; Dlubac et al., 2013; Kendrick et al., 2021; Knight et al., 2016; Maurer and Knight, 2016; Parsekian et al., 2015). These studies examined the widely used Schlumberger-Doll Research (SDR) model, an empirical equation that estimates K from NMR-derived porosity and transverse relaxation time (T_2), typically involving one to three sites (geographically located in the USA and Canada), a limited number of boreholes, and mainly focusing on inorganic geological materials such as sediments, with no consideration of units containing significant organic matter. Moreover, these studies did not consider geological information as a contributing factor in their data analysis; for example, they did not investigate whether different geological units, which potentially vary in chemical composition, physical properties, and pore structure, adhere to the same K estimation model. This study builds upon these works by gathering large datasets from multiple sites and many boreholes, focusing on a range of unconsolidated geological materials – including both organic and inorganic units – to address several key questions concerning the application of the SDR model for K estimation, including: (I) Is general calibration of the SDR model sufficient for reliable K estimation in unconsolidated geological materials? (II) Is geology-specific calibration beneficial for improving K estimation? (III) Is site-specific calibration a useful strategy for enhancing K estimation accuracy? (IV) Can a simpler SDR-based model be effectively used for K estimation in unconsolidated materials? And (V) if geological composition affects model performance, can specific geological material groups be unified to enable a successful calibration? To address these questions, we conducted laboratory and in situ field measurements. The laboratory dataset comprises NMR data of 92 samples taken from various geological units across nine sites in Denmark and Germany. For each laboratory sample, we collected NMR data, falling head K measurements, and volumetric magnetic susceptibility. For inorganic sediment samples, grain size distribution and gravimetric porosity were also measured. Including magnetic susceptibility and gravimetric porosity allows assessment of magnetic susceptibility influence on NMR-derived porosity and relaxation times, which can subsequently impact K estimation. The field dataset comprises 198 screened intervals from 128 boreholes located at eight sites in Denmark and Germany, spanning various organic and inorganic unconsolidated geological units. Each screened interval includes BNMR measurements and K values obtained from slug tests. The availability of independent K measurements enabled us to evaluate the uncertainty in K estimation as well as uncertainty in the underlying coefficients of the SDR model through a bootstrapping approach.

2. Theoretical background

2.1. Nuclear magnetic resonance (NMR)

In geophysics, NMR is used to detect hydrogen nuclei within porous media; specifically, within liquid water in the case of this work. The NMR signal originates from the bulk magnetization produced by hydrogen nuclei when immersed in a background magnetic field, which can be measured using a radiofrequency excitation pulse. This magnetization decays as hydrogen nuclei return to equilibrium in a static magnetic field through relaxation processes. The initial signal amplitude is proportional to the total number of hydrogen nuclei, which can be used to determine water content and/or porosity in case of full-saturation while the rate of signal decay over time defines a parameter called relaxation time (Coates et al., 1999; Dunn et al., 2002). The relaxation time in the directions parallel to and perpendicular to the polarizing field are defined separately as longitudinal (T_1) and transverse (T_2) relaxation times, respectively (Kenyon, 1997). Most often in geophysical application, the relaxation rate T_2 is measured for pragmatic reasons related to measurement speeds. It consists of three different relaxation mechanisms:

$$T_2^{-1} = T_{2B}^{-1} + T_{2S}^{-1} + T_{2D}^{-1}. \quad (1)$$

T_{2B} is the transverse relaxation time of the bulk fluid, and depends on pore fluid properties such as viscosity, temperature, and the presence of dissolved ferromagnetic and/or paramagnetic ions (Coates et al., 1999; Dunn et al., 2002). T_{2S} denotes the contribution of surface relaxation and depends on fluid – pore wall interactions including pore surface to volume ratio (S/V) and surface relaxivity (ρ_2), a material parameter accounting for paramagnetic impurities on the pore surface. T_{2D} is the so-called diffusion relaxation time that depends on the degree of heterogeneity in the static magnetic field. Assuming relaxation occurs in the fast-diffusion regime (Brownstein and Tarr, 1979), where the time-scale for the collision of hydrogen nuclei with the grain surface is much smaller than the time-scale for relaxation, surface relaxation can be explained by the following equation (Coates et al., 1999):

$$T_{2S}^{-1} = \rho_2(S/V), \quad (2)$$

The strong dependency of T_{2S} to pore surface to volume ratio forms the theoretical basis for estimating permeability and hydraulic conductivity, which is discussed in detail in the next section.

2.2. Hydraulic conductivity

Darcy's law describes the flow of water through a porous medium as (Freeze and Cherry, 1979):

$$Q = -KA \left(\frac{dh}{dl} \right), \quad (3)$$

Where Q is the flow rate, A is the cross-sectional area, dh/dl is the hydraulic gradient, and K is the hydraulic conductivity. Given that Q has units of volume per time and A has units of area, thus K is characterized by velocity units (e.g., m/s).

Hydraulic conductivity is different from intrinsic permeability (k) and is defined by:

$$K = \frac{k\rho g}{\mu} \quad (4)$$

where ρ is fluid density, the parameter μ is the dynamic viscosity of the fluid, and g is the gravitational acceleration constant. Note that k has units of area (e.g., m^2). While hydraulic conductivity depends on both the properties of the porous medium and the fluid (Equation (4)), permeability is an intrinsic property of the medium alone (Freeze and Cherry, 1979).

2.2.1. Laboratory and field measurements of K

K can be measured directly or estimated by empirical relationships using both laboratory and field-based methods. In the laboratory, measurements are made using constant or falling head tests, based on Darcy's Law (Freeze and Cherry, 1979; Woessner and Poeter, 2020). In the constant head method, water flows through the sample at a fixed rate under a steady hydraulic gradient. K can be calculated from the measured discharge and the known dimensions of the sample (Freeze and Cherry, 1979). In the falling head method, water is introduced into a standpipe connected to the sample, and the hydraulic gradient changes dynamically as the water level declines over time. By tracking the decrease in head and applying the appropriate equations, K is determined (ASTM, D., 2007). Constant head tests are useful for coarse-grained materials with high K values and falling head tests are best suited for fine-grained materials with low K values (Klute, 1965). While laboratory-based K measurements are often more accurate, the main limitation of these tests is the need for sampling and sample preparation, which can increase the time and cost of such measurements (Klute and Dirksen, 1986). Moreover, the small sample volumes used in these tests often struggle to capture the heterogeneity in the field, potentially leading to inaccuracies when extrapolating to larger scales (Freeze and Cherry, 1979).

In the field, in situ measurements of K are made at various scales, from the large-scale pumping tests (Freeze and Cherry, 1979) to the smaller-scale infiltration tests in unsaturated zones (Angulo-Jaramillo et al., 2000) and slug tests in saturated zones (Butler, 2019). Pumping tests involve long-term groundwater extraction, during which the drawdown of the water table is measured in the same well and/or in separate observation wells (piezometers). While pumping tests provide valuable K data, they are costly and can be problematic when dealing with polluted aquifers or aquifers with small yields. Additionally, pumping tests require a significant amount of time to perform and may not be practical in all settings. Slug tests, which are widely used in field applications, require borehole installation and development. In this method, a sudden change in water level, i.e., a "slug", is introduced into the well, and the recovery rate is measured. Slug tests provide an average K for the entire screened interval of the aquifer, and the estimated K value is most influenced by the area surrounding the wellbore (Butler, 2019). This method has quicker implementation and lower cost compared to pumping tests, but is limited in terms of scale and accuracy (Black, 2010; Butler, 2019). In unconsolidated formations, Hydraulic Profiling Tool (HPT) and Cone Penetration Test (CPT) can also provide field-based K measurement (Elhakim, 2016; McCall and Christy, 2020). The HPT method involves injecting fluid into a borehole while measuring the pressure response, which provides a profile of K at different depths (McCall and Christy, 2020). The CPT method uses a penetrometer to push a sensor into the ground, measuring the resistance of the soil to penetration and providing valuable data on subsurface characteristics, including K (Elhakim, 2016). These methods are useful for providing high-resolution data but are limited to a smaller range of soil types and penetration depth (Lunne et al., 2002; McCall and Christy, 2020).

2.2.2. K estimation

In addition to direct measurements, K can be estimated using transfer functions – empirical equations based on relationships between K and parameters that characterize porous material. These functions provide valuable insights when direct measurements are not feasible due to time, cost, or other logistical constraints. The following subsections cover the K estimation methods addressed in this paper.

2.2.2.1. Kozeny-Carman equation. The Kozeny-Carman equation is a widely used theoretical framework for estimating K . The equation was developed using a simplified model of several parallel capillary tubes with equal lengths and diameters to describe the packed bed and

simulates the pressure drop for laminar flow through a packed bed of solid grains (Rehman et al., 2024). The relationship predicts K from porosity (ϕ_r) and total pore surface area to volume ratio (S/V) (Gueguen and Palciauskas, 1994):

$$K = a_k \frac{\phi_r}{(S/V)^2} \quad (5)$$

a_k is a dimensionless constant coefficient factor that captures the tortuosity of the pore space and shape of the pores. The model assumes laminar flow in homogeneous, interconnected pore networks, making it effective for granular soils (e.g., sands) and intact rocks with uniform pore structures (Chapuis and Aubertin, 2003). Generally, the equation works well, but it struggles with structured soils (e.g., clays) and dual-porosity systems (e.g., fractured shale) due to non-uniform S/V distributions (Rehman et al., 2024).

2.2.2.2. NMR-based methods. The relationship between NMR relaxation times and pore characteristics is a key advantage of NMR data for the characterization of porous water-saturated material. Surface relaxation, one of the primary relaxation mechanisms, is particularly valuable as it encodes information about the pore surface to volume ratio (S/V) – a critical parameter for estimating hydraulic conductivity or permeability in saturated geological materials (compare equations (2) and (5)).

In standard T_2 measurements of geological formations, bulk relaxation (T_{2B}) is typically negligible compared to surface and diffusion relaxation mechanisms, unless pore fluids contain high concentrations of paramagnetic ions. Diffusion relaxation effects are usually minimized by optimizing the implemented measurement layout in order to work in conditions as magnetically homogeneously as possible. Consequently, T_2 relaxation times are often dominantly controlled by surface relaxation in most cases. Under these assumptions and if ρ_2 (equation (2)=) remains relatively constant in the entire investigated sample, variations in T_2 relaxation times directly reflect changes in surface to volume ratio of the pore:

$$T_2^{-1} \propto (S/V) \quad (6)$$

This relationship enables the estimation of properties such as permeability (k) or hydraulic conductivity (K), depending on the parameter of interest. To estimate K , the Kozeny-Carman equation (equation (5)) is adapted by replacing ϕ_r with NMR-derived porosity (ϕ) and S/V with the mean logarithm of T_2 (i.e., T_{2ML}). This results in the most widely used NMR- K model called the Schlumberger-Doll Research (SDR) model (Kenyon et al., 1988):

$$K_{NMR} = b\phi^m (T_{2ML})^n \quad (7)$$

where b , m , and n are empirical coefficients. Total porosity (ϕ) is expressed in fractions (dimensionless), T_{2ML} is in seconds, and b has units of (m/s^{n+1}), resulting in K_{NMR} in meters per second (m/s). While the Kozeny-Carman framework suggests theoretical exponents $m = 1$ and $n = 2$, empirical calibration by Kenyon et al. (1988) demonstrated that $m = 4$ and $n = 2$ yield optimal permeability estimates. These values are standard in petroleum NMR logging applications (Kenyon, 1997). If independent measurements of K are available, the coefficients b , m , and n can alternatively be determined through empirical curve fitting (Maurer and Knight, 2016). Accordingly, we treat b , m and n as free fitting parameters to allow a certain degree of flexibility in our study, and to achieve a higher K estimation accuracy. Other NMR-based K estimation models exist but are not as widely adopted as the SDR model (Coates et al., 1999).

Kenyon (1997) highlights that effective permeability estimation via NMR requires accounting for pore throat-size ratios, which govern pore connectivity. In other words, a good correlation between pore body and pore throat sizes must exist for the SDR model to work (Freedman, 2006). To address this concern, researchers have integrated SDR model

with auxiliary datasets such as magnetic susceptibility (Crow et al., 2021), and gamma ray measurements (Daigle and Dugan, 2009). Geological context is critical here: depositional environments and diagenetic processes dictate pore structures (e.g., pore throat to body size ratios) and mineralogical compositions that influence surface relaxivity. As a result, the constant coefficients of the SDR model are expected to be geology dependent (Maurer and Knight, 2016). The availability of geological data in our dataset provides an opportunity to evaluate if the constant coefficients of the SDR model are different for various unconsolidated geological units.

An alternative approach for improving K estimation from NMR data is to calibrate based on the field site location (i.e., a site-specific calibration). The entire dataset collected at a single site are commonly used to calibrate the SDR model (e.g., Dlubac et al. (2013)); this approach is effectively a site-specific calibration. In this study, data is available from multiple sites which makes it possible to evaluate if site-specific calibration is useful.

Many research studies on NMR-derived K estimations for unconsolidated geological materials observed that porosity does not contribute to K estimation, at least for some of their datasets (Kendrick et al., 2021; Maurer and Knight, 2016). For instance, Maurer and Knight (2016) repeatedly observed that porosity had no influence on K estimation for all the models they investigated, including SDR. Mashhadi et al. (2024) also reported $m = 0$ for the geology-specific calibrations of the SDR model for sand and gytja in their dataset. Thus, it seems reasonable to exclude porosity and consider a purely empirical model for unconsolidated geological materials, we suggest to call it “simplified SDR” model:

$$K_{NMR} = b(T_{2ML})^n \quad (8)$$

Note that Yaramanci et al. (1999) also used this format of empirical formula to estimate K in unconsolidated geological units in their study area from surface NMR data. In this study (Sections 5.4 and 5.5), we will provide further evidence to evaluate the effectiveness of this simplified model for unconsolidated sediments.

2.3. Statistical analysis

2.3.1. NMR- K calibration & estimation error metrics

To calculate the optimum constant coefficients of the SDR model, the log transformation of the model is considered to make the problem linear:

$$\log_{10}(K_{NMR}) = \log_{10}(b) + m\log_{10}(\varphi) + n\log_{10}(T_{2ML}) \quad (9)$$

This form has two advantages: (1) it simplifies the problem and makes inversion more stable, (2) it minimizes the error for both small and large K values.

A simple linear curve fitting is applied to determine the coefficients in equation (9). The fit is evaluated with the standard R^2 score, which determines how much of the variance in the data can be explained by the model. To assess error, we use the K_{diff} metric proposed by Kendrick et al. (2021):

$$K_{diff} = \text{mean}\left(\max\left(\frac{K_{NMR}}{K}, \frac{K}{K_{NMR}}\right)\right) \quad (10)$$

This metric quantifies the average deviation of the estimated hydraulic conductivity (K_{NMR}) from the measured value (K), emphasizing deviations irrespective of over- or under-estimation. In practice, knowledge of K within one order of magnitude is very useful in hydrogeological applications (Freeze and Cherry, 1979); thus, it is a commonly accepted criterion that the estimated value (i.e., K_{NMR} in our study) should lie within one order of magnitude of the true measured K value. Accordingly, for a single point, this corresponds to a $K_{diff} < 10$. This threshold is consistently applied throughout our data analysis and interpretation.

Both the R^2 score and K_{diff} are useful in ensuring the reliability of model performance. A high R^2 score indicates that the model represents the data, i.e., it follows the SDR model. The K_{diff} measures the mean error in K estimations. A low R^2 value suggests that the model explains only a small proportion of the variance. However, if the K_{diff} is also low, it indicates that prediction errors are minimal. This scenario implies that, despite the model's limited ability to explain variance, the low variability in the data results in small estimation errors, allowing the model to remain useful for predictive purposes. If the R^2 score is too small (e.g., $R^2 < 0.25$), the data does not follow the K estimation model, hence constant coefficients (i.e., b , m , and n) and their distributions will not be valid.

Finally, to visually assess the calibration outcomes and SDR model performance, a “validation plot” is generated by plotting the K_{NMR} against the measured “true” K values for each calibration scenario. This provides a clear and intuitive evaluation of how well the model reproduces the true K values. In addition, we also calculated the percentage of points for which the estimated K_{NMR} lies within one order of magnitude of the measured K , denoted as PER_{1OM} . This metric complement K_{diff} and can be useful in cases where a few extreme outliers heavily influence the K_{diff} . However, PER_{1OM} should be interpreted with caution, as it may overstate model performance – particularly when R^2 is low – and should always be considered in conjunction with both R^2 and K_{diff} .

2.3.2. Bootstrapping

Bootstrapping is a statistical method that involves repeatedly resampling the original dataset – with replacement – to create a large number of “bootstrap samples”. These samples allow for the empirical estimation of the distribution of almost any statistic, thereby enabling the calculation of standard errors and confidence intervals (CI) without reliance on traditional parametric assumptions. This nonparametric approach is especially valuable when the theoretical distribution of the estimator is unknown or complex, providing a flexible and practical alternative for interval estimation. The number of bootstrapping iterations depends on the purpose of the study, but 2000 iterations, as is used in this study, are typically sufficient for stable estimates of standard errors and basic confidence intervals (Efron and Tibshirani, 1994).

To perform the complete statistical analysis, a table of b , n , m , R^2 , PER_{1OM} and K_{diff} is created with the bootstrapping approach. These values are then plotted as histograms. From the histogram, the median, which is preferred over the mean due to its robustness against outliers, and the 95 % CI are determined. The 95 % CI is defined by the 2.5th percentile ($Q_{0.025}$) also known as the Lower Confidence Limit (LCL) and the 97.5th percentile ($Q_{0.975}$) or Upper Confidence Limit (UCL).

3. Data collection

3.1. Laboratory measurements

The full set of samples were placed in two categories: glacial sediment samples, consisting of inorganic unconsolidated sediments such as sand, sandy till, clay till and clay; and organic samples, which include peat soils. Note that geological information was obtained from the available borehole geological logs, such as those provided by Jupiter (Denmark's national borehole database). For the most part, the measurements collected on the sediment and organic samples were the same (details follow), however, the sampling procedure was different. The organic samples were collected trying to preserve the in-situ conditions while sediment samples are dried, disaggregated and then analyzed.

Sediment samples (inorganic) were collected from nine sites in Denmark and Germany (Fig. 1). NMR measurements (T_2 CPMG), K , volumetric magnetic susceptibility, and gravimetric porosity were measured for all sediment samples. The details of sample preparation, NMR and magnetic susceptibility measurements for sediment samples are described in detail in Mashhadi et al. (2025). For the inorganic

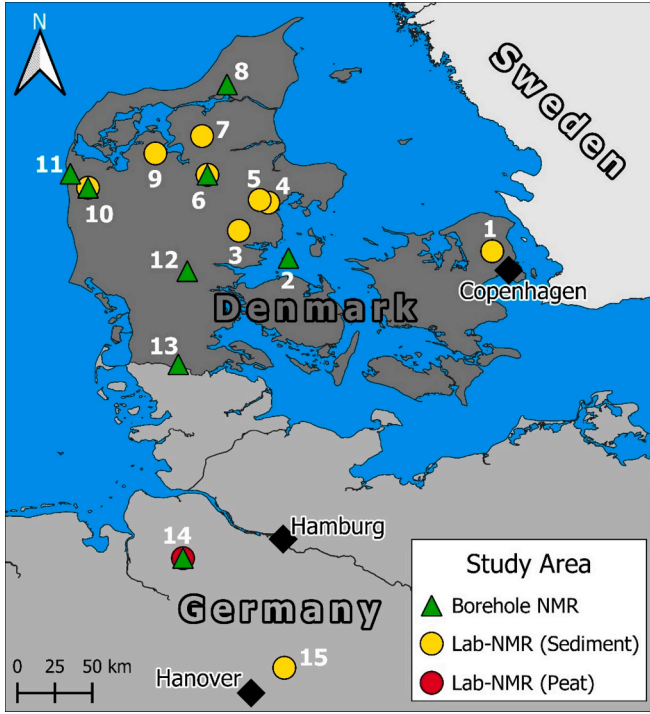


Fig. 1. Location map of study areas where field measurements (i.e., borehole NMR and slug tests) were conducted and where unconsolidated geological samples were collected for laboratory measurements (e.g., Lab NMR). The sites are numbered on the map as follows: 1. Farum, 2. Endelave, 3. Horndrup Bak, 4. Kasted, 5. Hinnerup, 6. Vejrumbrø, 7. Oddebak Himmerland, 8. Ringfæner, 9. Hagens Mollebak, 10. Norre Vosborg, 11. Bovling, 12. Runkenbjerg, 13. Jyndevad, 14. Gnarrenburg, 15. Schillerslage.

sediment samples, gravimetric porosity was also measured, which enables us to validate whether NMR data can accurately estimate porosity and to evaluate whether magnetic susceptibility contributes to porosity estimation errors in the studied samples. K values were also recorded using the falling head approach (ASTM, D., 2007). Note that grain size distribution data was also collected for these sediments (refer to Appendix A for more details).

Organic samples were collected from a peatland site near Gnarrenburg (Gnarrenburger Moor) in Northwest Germany (Fig. 1). The area, a former bog, was drained for agricultural use decades ago and has since been used as grassland that is regularly mowed for feeding livestock. Samples were taken at various depths from 5–50 cm and exhibit a wide range of decomposition degree. Their organic content varies between 90 % and almost 100 %. To obtain sample material as undisturbed as possible in order to preserve its original hydraulic properties and to allow NMR measurements at the same time, plastic sample rings (250 ml) were hammered into and afterwards carefully removed from the soil. Afterwards, they were sealed and stored in a cooling box (4 °C) for transport to the laboratory, where NMR and K were measured. Additional sample material next to the sampling ring positions were collected in a plastic bag to determine their magnetic susceptibility. In the lab, the samples in the plastic rings were saturated from bottom to top. The NMR measurements were performed using a core scanner (Helios device, Vista Clara) working at the same frequency as the borehole equipment used in this study. The T_2 relaxation was investigated with an echo time of 200 μ s, while the number of echoes was adjusted between 1000 and 1500 depending on the individual relaxation rate. To increase the data quality, the measurement repetitions (number of scans) were set to at least 32 to achieve signal-to-noise (SNR) ratios between 20 and 50. After the NMR measurement, K was measured using the falling head method (KSAT device, METER Group). The volumetric magnetic susceptibility of the organic samples was measured analogous to the inorganic samples

using the MS2 device of Bartington® instruments (Mashhadi et al., 2025).

3.2. Field measurements

Field measurements were collected across different geological units at eight sites in Denmark and Germany (Fig. 1). Borehole NMR data and slug test measurements were collected at each borehole; the dataset consists of 198 screens from 128 boreholes. Geological units include organic (peat, and gyttja) and inorganic (sand, clay) materials. Geological data is sourced from the available borehole geological logs (e.g., Jupiter database).

Borehole NMR data were collected using a Dart DP140N logging probe from Vista Clara and a standard T_2 CPMG sequence, at a vertical resolution of 22 cm. The measurement parameters were: 500 μ s echo time, 1000 echoes, 3 s inter-experiment delay time, and 16 stacks. Data were inverted using the Javelin Pro Plus software (Vista Clara).

Slug test data were collected with pressure transducers and then analyzed to obtain K values according to Zlotnik et al. (2010) for unconfined geological units and Hvorslev (1951) for confined units. Where available, data were sourced from prior measurements available through project collaborators. In the investigated boreholes, screen length ranges from 30 cm to 3 m.

A database was created to combine BNMR data with K data for each screened interval. As the screen length was variable and often larger than the vertical resolution of BNMR data (see Fig. 2), a representative value of NMR parameters (e.g., total porosity, and T_{2ML}) was defined for each screened interval. A weighted average based on the overlapping length is employed. The calculated weighted average for a BNMR parameter (f_{BNMR}) is determined by considering the contributions from individual BNMR measurements that overlap with the screen:

$$f_{BNMR} = \frac{\sum_{i=1}^{i=n} l_{BNMRi} f_{BNMRi}}{\sum_{i=1}^{i=n} l_{BNMRi}} \quad (11)$$

Here, f_{BNMRi} represents the value of the i^{th} BNMR measurement with overlap, and l_{BNMRi} denotes the length of the sensitive zone that overlaps with the screen.

Throughout our analysis, several factors should be considered when

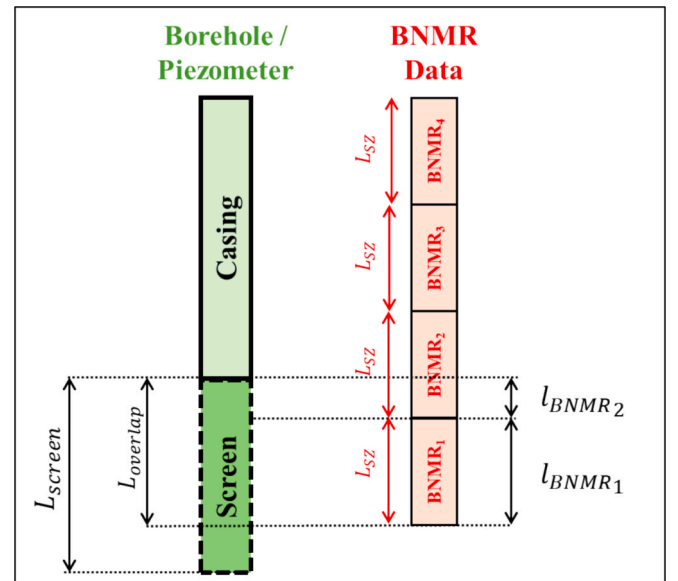


Fig. 2. Schematic illustration of a shallow borehole/piezometer, showing the parameters used to calculate the average BNMR values and screen coverage (L_{sz} : length of sensitive zone or vertical resolution of the logging probe, and $BNMR_i$ is the i^{th} borehole NMR measurement).

using slug test data to calibrate NMR-based K estimation models. One important factor, which determines the extent of overlap between the BNMR measurements and the screen, is called “screen coverage”. It represents the percentage of screened interval that is covered by BNMR measurements and is calculated by the equation below (Fig. 2):

$$\text{Screen coverage} = \frac{L_{\text{overlap}}}{L_{\text{screen}}} \times 100 \quad (12)$$

Screen coverage is crucial because the hydraulic conductivity obtained from slug test is influenced by the entire screened interval. If the screen coverage is insufficient, such that the BNMR data cannot capture all pores affecting K , expected correlations may not be observed, particularly if the characteristics of the porous media vary within the depth range of the screened interval.

In our dataset, only 48 screens out of 198 have full screen coverage. This problem can be common and inevitable in shallow boreholes and piezometers with small screen lengths, particularly when the borehole/piezometer end is screened. Since the sensitive zone of BNMR logging probes is usually located a certain distance from the bottom of the probe (e.g., the sensitive shell of the Dart140N logging probe starts 8 in. from the bottom), BNMR data cannot be measured for a portion of the deepest part of the borehole (e.g., approximately 20 cm with the Dart140N probe), resulting in poor screen coverage. In some cases, only a single BNMR measurement may partially overlap the screened interval, which limits its usefulness for calibration purposes. Additionally, the possibility of clay or sand accumulation at the borehole bottom, caused by material entering the borehole from the aquifer, can further reduce screen coverage. We considered a minimum screen coverage of 50 %, resulting in the exclusion of 30 data points (screens) from the field dataset. This threshold is a preliminary choice to retain enough data points in the dataset for calibration and bootstrapping analysis purposes while removing outliers, thereby making the results more meaningful.

Apart from screen coverage, a minimum value for $l_{\text{BNMR}i}$ is also considered. Setting a minimum threshold for $l_{\text{BNMR}i}$ ensures that only measurements with sufficient overlap contribute to the weighted mean calculations, incorporating those that are predominantly influenced by the porous media within the screened interval rather than outside of it. This can be important when the top or bottom of the screen is adjacent to geological boundaries with distinct pore structures and hydrogeological properties. Thus, in our weighted average calculations, we only considered those BNMR measurements for which at least half of the

measured depth interval (i.e., the sensitive zone) overlapped with the screen (i.e., $l_{\text{BNMR}i} \geq 11\text{cm}$). A total number of 167 screens remained in the dataset for subsequent analysis.

4. Results

4.1. Laboratory data analysis results

4.1.1. Magnetic susceptibility effects on NMR data

For organic samples (i.e., peat soils), the absolute values of magnetic susceptibility were very small, ranging from 1.1×10^{-6} to 3.1×10^{-6} in SI units. Notably, 11 out of 19 peat samples exhibited negative magnetic susceptibility values. Such negative values have also been reported for peat soils by Garcés-Pastor et al. (2023) among others. The weak magnetic properties of these peat samples do not seem to have any noticeable influence on relaxation times.

In inorganic samples (i.e., sediments), a range of magnetic susceptibility values are observed that spans several orders of magnitude. As discussed in detail by Mashhadi et al. (2025), there is only a weak negative correlation between relaxation time (T_{2ML}) and magnetic susceptibility (Fig. 3a), suggesting that relaxation times are slightly affected by magnetic susceptibility and other factors – such as pore size – have a dominant control on T_{2ML} variations. The NMR-derived porosity to gravimetric porosity ratio vs. magnetic susceptibility plot shows that NMR-based estimation of porosity is hugely influenced by magnetic susceptibility (Fig. 3b). A significant underestimation happens for high magnetic susceptibility values, i.e. up to the maximum of 70 % underestimation in our set of samples. This typically happens when magnetic susceptibility exceeds $\sim 10^{-4}$ SI units, consistent with the findings of Keating et al. (2020). As Fig. 3b illustrates, the porosity ratio remains relatively stable below this threshold, while a sharp decline occurs above it, indicating increasing NMR porosity underestimation with higher magnetic susceptibility. This trend is further supported by Pearson’s correlation results, which shows statistically significant negative correlations (i.e., with $p < 0.05$) between magnetic susceptibility and porosity ratio in clay till ($r = -0.75$), sand ($r = -0.80$), and clay ($r = -0.998$). Note that the sandy till also showed a weaker but statistically insignificant correlation, and the clay result is based on only three samples.

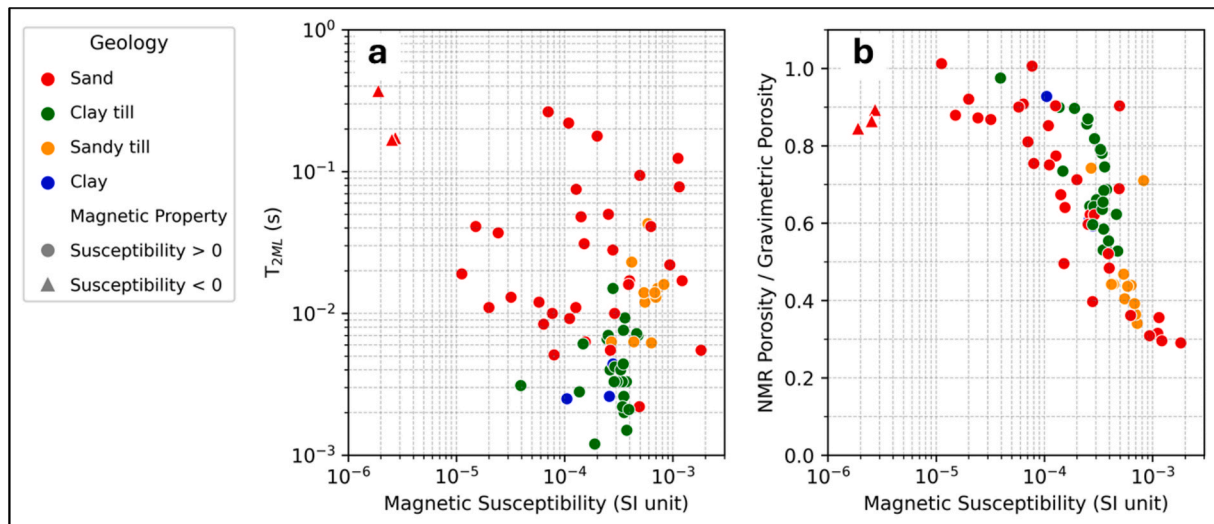


Fig. 3. T_{2ML} (a) and NMR-derived porosity to gravimetric porosity ratio (b) vs. absolute value of volumetric magnetic susceptibility plotted color coded by geological composition. These data are related to the laboratory dataset. Note that drying technique for gravimetric porosity determination may lead to overestimation or underestimation of true porosity, especially for clay-rich samples (Fleury et al., 2022). That can be a reason for observing ratios smaller than 1 for samples with small magnetic susceptibility values.

4.1.2. K estimation by laboratory NMR data

4.1.2.1. General calibration. We first consider a general calibration based on calibrating the entire laboratory dataset simultaneously. Before curve fitting, it is insightful to plot total porosity and T_{2ML} vs. falling head K measurements of laboratory samples (Fig. 4). Organic soils such as peat have higher porosity values compared to mineral sediments, as seen for peat soils from the Gnarrenburg site (Fig. 4a). Considering the entire dataset, there seems to be no obvious correlation between total porosity and K values. On the other hand, the T_{2ML} vs. K plot indicates that there seems to be a correlation between these parameters (Fig. 4b). Note that some points appear to deviate from this general trend, especially the points related to the peat samples from the Gnarrenburg site.

Performing the curve fitting for the entire laboratory dataset to calculate the best constant coefficient values of SDR model results in b , n , and m values of 0.028, 1.07 and 1.72, respectively, with an R^2 score of 0.57, indicating more than half of the variance is explained by the model. The K_{diff} equals 11.7, which is larger than the acceptable threshold. With a PER_{10M} of 77.2 %, it is evident that a substantial number of points still falls outside the accepted error margin.

4.1.2.2. Geology-specific calibration. Geology-specific calibration is performed using the data points in each geological unit independently. To support this approach, correlations between NMR parameters and K were evaluated for each unit (Fig. 4). Total porosity did not have a significant correlation with K for any of the geological units except clay, which includes only three data points (Fig. 4a). In contrast, T_{2ML} and K were correlated for some of the geological units such as sand, and sandy till (Fig. 4b). There was no strong correlation between T_{2ML} and K for clay and clay till, and the peat geological unit shows obviously no correlation.

The geology-specific calibration results indicate successful K estimation for sand, sandy till, and clay and failure of the SDR model to provide a good K estimation for clay till and peat (Table 1 and Fig. 5b). The K estimation accuracy is improved in successful cases with K_{diff} of 4.3, 2.6, and 1.0 for sand, sandy till, and clay, respectively (with $PER_{10M} > 90$ %). The successful geology-specific calibrations are noteworthy since our dataset is gathered from multiple sites (e.g., sand geology samples are related to 8 different sites) meaning that NMR- K relationship holds regardless of the site-specific and/or pore structure variability for the considered samples. The lack of NMR data correlation with K for the peat soil and thus unsuccessful K estimation has also been previously reported by Mashhadi et al. (2024) in a borehole NMR study. For the peat unit, the absence of significant correlation between both T_{2ML} and

porosity with K causes the regression coefficients m and n to approach zero, effectively reducing the SDR model to a constant prediction. This results in predicted values clustering tightly around the average K , forming a pattern resembling a horizontal line in the validation plot.

4.1.2.3. Site-specific calibration. Site-specific calibration is performed independently for each site to examine whether this approach can reliably estimate K . With the exceptions of Gnarrenburg, Farum and Horndrup Bak sites, other sites show a successful K estimation with high R^2 scores, and very small K_{diff} values – often below 3 – which corresponds to PER_{10M} values of 100 % in many cases (Fig. 5c; Table 1). At the Farum and Horndrup Bak sites, R^2 scores are relatively low, likely due to the limited range of T_{2ML} and/or K values (spanning only about one order of magnitude), which reduces the quality of the regression fit. However, the K_{diff} remains low at these sites, suggesting that the K estimations are still acceptable. In contrast, the Gnarrenburg site, from which the peat soil samples originate, does not suffer from limited variability in NMR parameters, yet site-specific calibration fails to produce reliable K estimates, as indicated by a poor R^2 score and a high K_{diff} value. It is worth noting that some sites have very limited datasets (e.g., Norre Vosborg with only four datapoints), which is not ideal for statistically robust calibration. Nevertheless, the overall performance of site-specific calibrations of the SDR model appears promising across most sites.

4.2. Field data results

4.2.1. Slug test measurements

The K values for the screened intervals range from 2.18×10^{-9} to 1.01×10^{-3} m/s (Table 2). We observe high variability of K values within each geological unit; for example, K for sand varies over six orders of magnitude and K for peat varies over 4 orders of magnitude. The label “mixed” represents screens that intersect multiple geological units, e.g., sand and clay, sand and peat; K for the mixed geological units has a similar range as the sand units.

4.2.2. K estimation by borehole NMR data

4.2.2.1. General calibration. Examination of NMR data vs K values in the field dataset highlighted similar trends to those observed in the laboratory dataset. There was no correlation between total porosity and K , while T_{2ML} showed a general linear trend with K (Fig. 6). The constant coefficients b , m , and n for the general calibration, i.e., using the entire field dataset to derive estimates of K , were 1.0×10^{-3} , 0, and 1.91, respectively. The R^2 score was 0.42 and the mean K_{diff} was 22.8,

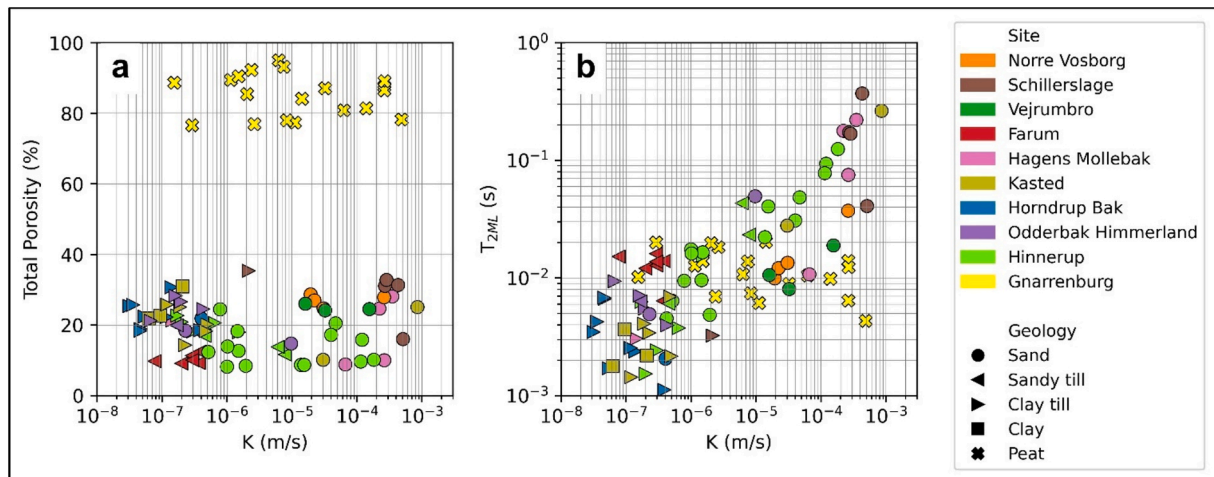


Fig. 4. Porosity (a) and T_{2ML} (b) measured by Lab NMR vs. K measured by falling head tests.

Table 1SDR coefficients (b , m , and n), R^2 score, K_{diff} and PER_{10M} resulting from different calibration approaches for the laboratory dataset.

	Geological Unit / Site	No. of data points	b	m	n	R^2	K_{diff}	PER_{10M} (%)
General Calibration	Entire Dataset	92	2.86×10^{-2}	1.074	1.72	0.57	11.7	77.2
	Inorganic Geological Units	73	2.93×10^{-2}	0.861	1.814	0.7	6.5	84.9
Geology-specific Calibration	Sand	35	1.32×10^{-2}	0.647	1.469	0.68	4.3	94.3
	Sandy till	11	3.19×10^{-2}	1.79	1.689	0.49	2.6	100
	Clay till	24	2.58×10^{-7}	0	0.058	0	3	95.8
	Clay	3	1.07×10^{-4}	3.196	0.406	1	1	100
	Peat	19	1.15×10^{-5}	0	0	0	15	57.9
	Hinnerup	23	3.62×10^{-3}	0	1.674	0.83	2.3	100
Site-specific Calibration	Schillerslage	5	2.35×10^{-3}	0	1.077	0.77	2.7	100
	Oddebak Himmerland	7	3.97×10^{-4}	0	1.479	0.61	2.5	100
	Farum	7	1.80×10^{-5}	1.89	0	0.14	1.5	100
	Hagens Mollebak	5	6.37×10^{-3}	0	1.53	0.76	4.6	80
	Kasted	10	9.04×10^{-3}	0	1.82	0.93	2.1	100
	Norre Vosborg	4	7.46×10^{-1}	1.007	2.032	0.99	1.1	100
	Vejrumbro	3	6.24×10^{-1}	0	2.155	0.66	1.7	100
	Hornstrup Bak	9	8.89×10^{-8}	0	0	0	2.5	100
	Gnarrenburg	19	1.15×10^{-5}	0	0	0	15	57.8

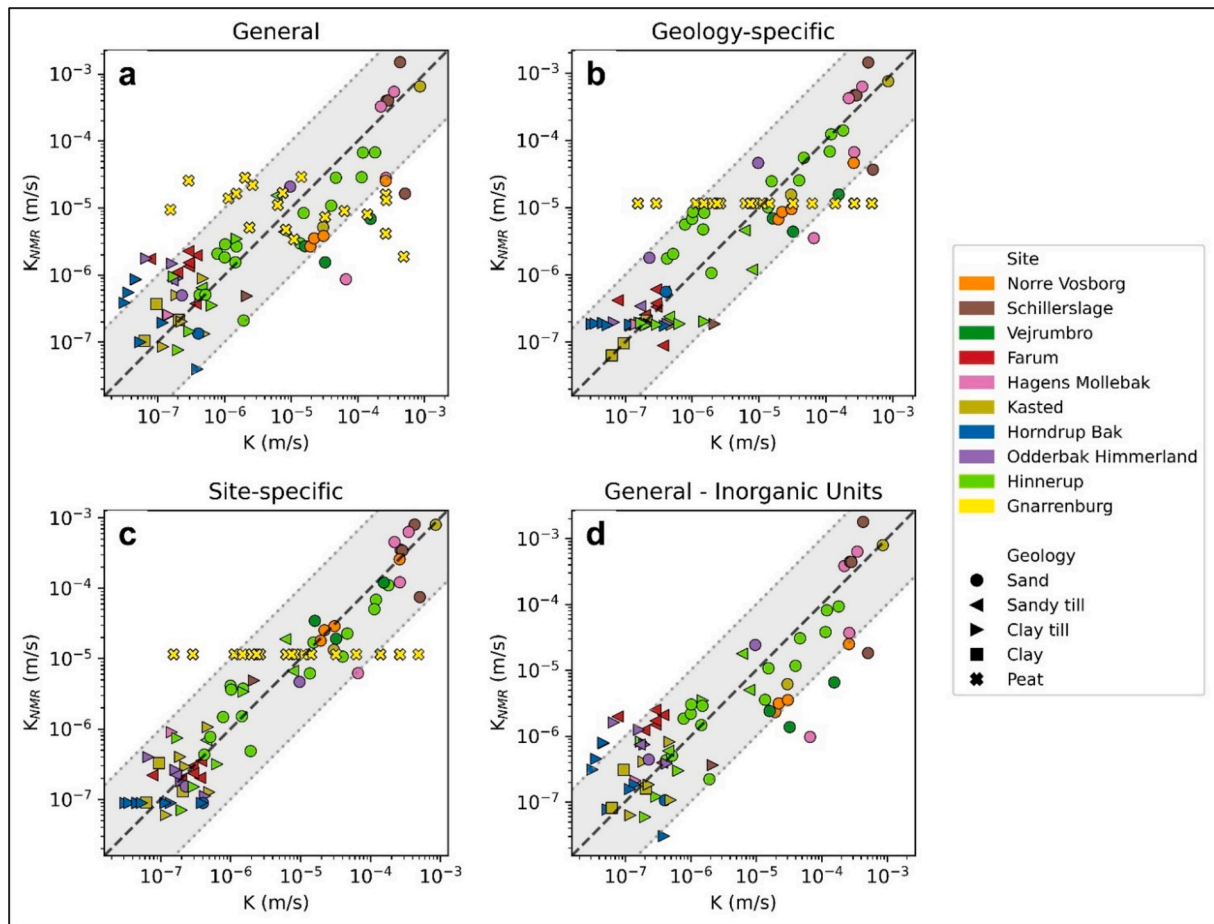


Fig. 5. Validation plots (K_{NMR} vs. K) for different calibration strategies applied to the laboratory dataset: (a) general calibration using all datapoints, (b) geology-specific calibration for each geological unit, (c) site-specific calibration for each site, and (d) general calibration for inorganic geological units. Each plot includes a dashed black line representing perfect agreement ($K_{NMR} = K$), along with two gray dotted lines and a shaded area that indicate the bounds and region within which K_{NMR} deviates by no more than one order of magnitude from K .

indicating that the SDR model performance for K estimation was not very good for the general calibration approach. This is further illustrated in the validation plot (Fig. 7a), where many data points fall outside the desired shaded area. The corresponding PER_{10M} value of 77.8 % confirms that the high K_{diff} is not merely driven by a few extreme outliers but reflects a broader mismatch across the dataset.

4.2.2.2. Geology-specific calibration. Plotting total porosity and T_{2ML} values obtained from borehole NMR measurements against slug test K values suggests that porosity did not correlate with K for any of the geological units (Fig. 6a). In contrast, T_{2ML} seems to correlate with K in sand, clay, and to some degree gyttja. Note that T_{2ML} of peat samples did not correlate with K .

The geology-specific calibration showed that the SDR model does not

Table 2
Hydraulic conductivity values from slug tests.

Geology	Number of measurements	K (m/s)	max	min
		median		
Sand	55	7.03×10^{-5}	1.01×10^{-3}	2.18×10^{-9}
Clay	9	6.24×10^{-7}	5.00×10^{-6}	1.04×10^{-7}
Peat	65	1.08×10^{-6}	1.25×10^{-4}	7.25×10^{-8}
Gyttja	9	5.55×10^{-6}	9.91×10^{-4}	2.26×10^{-8}
Mixed	60	4.84×10^{-6}	3.18×10^{-4}	1.01×10^{-9}

predict K for peat (Table 3 and Fig. 7b), as observed for the laboratory dataset and as reported in Mashhadi et al. (2024). Similar to the lab dataset results, the lack of correlation between T_{2ML} and porosity with K in peat causes the SDR coefficients m and n to approach zero, reducing the model to a constant prediction. This results in points clustered tightly around a horizontal line in the validation plot. For the clay units, K was well-estimated with a K_{diff} of 2.6; but the R^2 score was low mainly resulting from the limited T_{2ML} range. For the gyttja unit, although K_{diff} was 10, K still appeared to be reasonably estimated, with a high R^2 score and a validation plot showing all but two points estimated within one order of magnitude error (Fig. 7b). For the sand units, the R^2 score was low, and the K_{diff} was elevated (16.6), indicating a possible poor estimation. However, the validation plot highlights that most of the K values were estimated with acceptable error with just a few outlier points that negatively influenced the defined error metric, as highlighted by PER_{10M} of 82.5 %.

4.2.2.3. Site-specific calibration. Site-specific calibration led to both successful and unsuccessful cases of K estimation in the field dataset (Fig. 7c, Table 3). Bovling and Endelave showed very promising results in terms of K estimation with site-specific calibration; highlighted with high R^2 scores, very small K_{diff} values, and a PER_{10M} of 100 %. Jyndevad site-specific calibration is also acceptable as also highlighted in the validation plot (Fig. 7c); characterized by a small K_{diff} but a low R^2 score merely due to limited range of K values. In contrast, for the Norre Vosborg site, the R^2 score was 0.43 and K_{diff} was very high (65), with a PER_{10M} of 58.3 %. The K estimation resulted in a validation plot with multiple extreme outliers, including both severe underestimations and overestimations. Such a high number of outliers was unexpected, especially given that site-specific calibration efforts in mineral soils have generally been promising. We reviewed both the slug test and NMR measurements and found no obvious signs of inaccuracy or irregular noise, so the reason for the poor performance remains unclear. Sites such as Ringfennner, Vejrumbro, and Runkenbjerg, which include data from peat geological units, generally showed poor K estimation results,

marked by low R^2 and high K_{diff} values (Fig. 7c, Table 3). The site-specific calibration seems to work for Runkenbjerg with small R^2 and acceptable K_{diff} . However, the $T_{2ML} - K$ correlation at this site is primarily influenced by sand data while T_{2ML} and K values for peat cluster around the site's mean values – acting neutrally rather than exerting a problematic effect. This suggests that the moderately effective calibration at Runkenbjerg may be coincidental.

5. Discussion

5.1. Is general calibration possible?

Our results show that when a general calibration of the SDR model was used for unconsolidated geological materials inclusive of both organic and inorganic sediments, K was poorly estimated for both the laboratory and field datasets used in this study. These results are consistent with studies of rock cores, where it has not been possible to find a general calibration (Kenyon, 1997). The difficulty in obtaining a general calibration for rock cores is typically attributed to (1) the presence of paramagnetic ions (e.g., iron, manganese, nickel, and chromium) on the pore surface which increase the surface relaxivity (Dunn et al., 2002; Kenyon, 1997); (2) the presence of magnetic minerals which enhance diffusion relaxation (Dunn et al., 2002); (3) the fact that surface relaxation depends on pore body size and not pore throat size, whereas K is dominated by the pore throat size (Kenyon, 1997; Kleinberg and Vinegar, 1996); and (4) in large pores, e.g., gravels or vuggy pores found in carbonates, bulk relaxation may dominate and thus the relaxation time distribution may not reflect pore size distribution anymore (Kenyon, 1997; Kleinberg and Vinegar, 1996).

In our datasets (Figs. 5 and 7, Table 1 and 3), the difficulty in obtaining a general calibration for the SDR model arose from the inclusion of the organic units within the datasets, as when the organic units were excluded from the calibration (e.g., in the geology- or site-specific calibrations), the SDR model more accurately predicted K . These observations are consistent with a previous study where K could not be reliably predicted from NMR data in organic peat soils (Mashhadi et al., 2024). Costabel and Stange (2025) found that the surface relaxivity significantly varies among the samples from the Gnarrenburg site and strongly depends on the individual decomposition degree of the peat material. This might be true for most organic soils also from other sites and suggests that the NMR relaxation time of organic soils might not be an accurate and unambiguous proxy for their pore sizes as in inorganic soils and sediments. Instead, contrasts of T_2 in organic soils must be attributed to both a possible change of pore sizes but also to a possible change of pore surface properties resulting from chemical

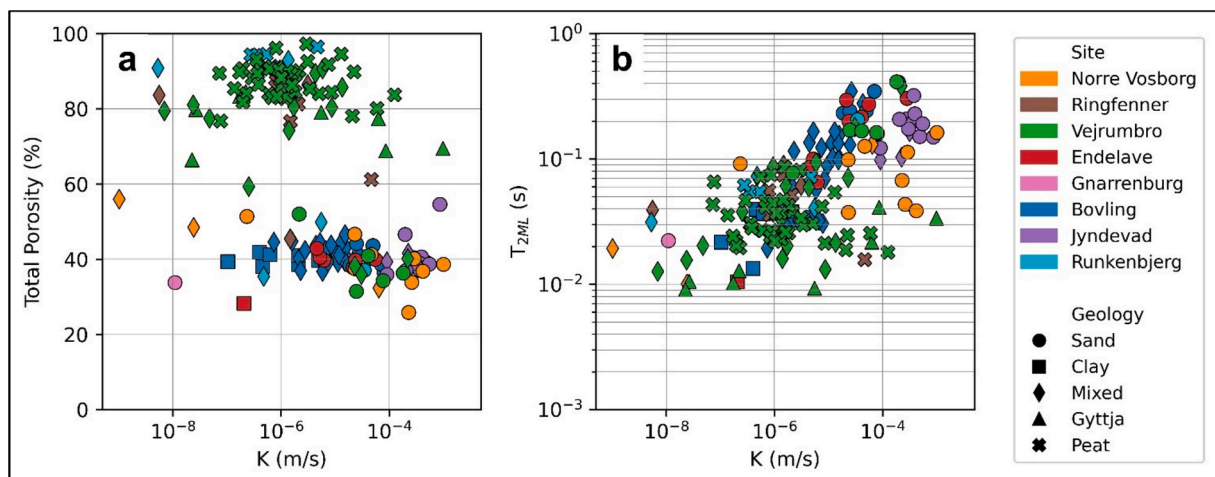


Fig. 6. Porosity (a) and T_{2ML} (b) measured by BNMR vs. K measured by slug tests.

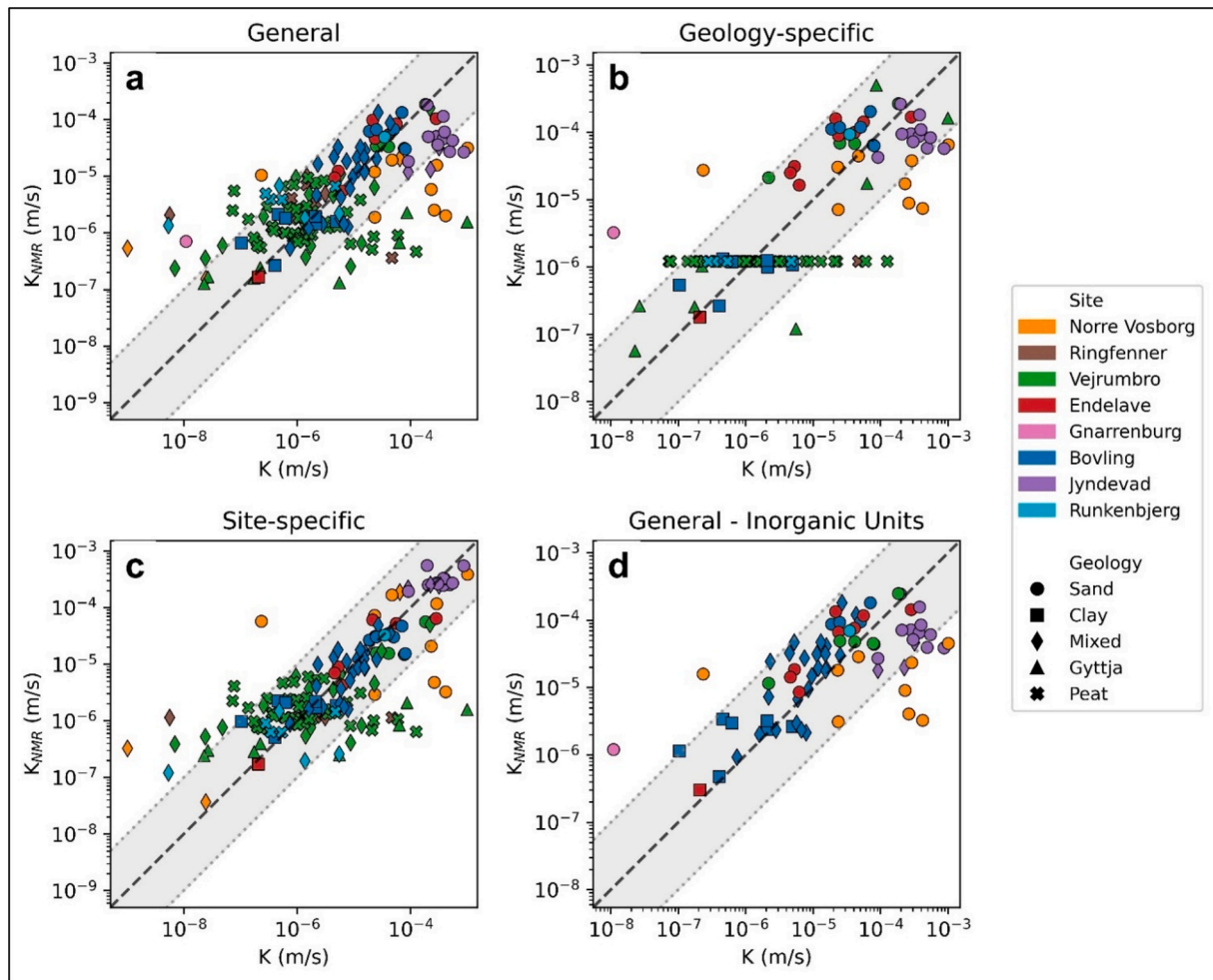


Fig. 7. Validation plots (K_{NMR} vs. K) for different calibration strategies applied to the field dataset: (a) general calibration using all datapoints, (b) geology-specific calibration for each geological unit, (c) site-specific calibration for each site, and (d) general calibration for inorganic geological units. Each plot includes a dashed black line representing perfect agreement ($K_{NMR} = K$), along with two gray dotted lines and a shaded area that indicate the bounds and region within which K_{NMR} deviates by no more than one order of magnitude from K .

Table 3

SDR coefficients (b , m , n), R^2 score, K_{diff} and PER_{1OM} resulted from different calibration approaches for the field dataset.

	Geological Unit / Site	No. of data points	b	m	n	R^2	K_{diff}	PER_{1OM} (%)
General Calibration	Entire Dataset	167	1.01×10^{-3}	0	1.911	0.42	22.8	77.8
	Inorganic Geological Units	75	1.259×10^{-3}	0	1.828	0.45	9.2	84
Geology-specific Calibration	Sand	40	1.01×10^{-3}	0	1.509	0.21	16.6	82.5
	Clay	8	1.77×10^{-4}	0.06	1.496	0.36	2.6	100
	Gyttja	8	3.00×10^{-5}	3.469	5.927	0.73	10	75
	Peat	58	1.21×10^{-6}	0	0	0	7	86.2
	Norre Vosborg	9	1.64×10^{-1}	0	3.328	0.43	65.4	58.3
Site-specific Calibration	Bovling	36	3.31×10^{-4}	0.491	1.401	0.67	2.3	100
	Endelave	9	5.17×10^{-4}	0	1.757	0.88	1.8	100
	Jyndeved	14	4.58×10^{-3}	2.335	0.37	0.24	1.6	100
	Ringfener*	9	1.15×10^{-6}	0	0	0	28.7	77.8
	Runkenbjerg*	9	3.62×10^{-3}	0	2.979	0.44	7.2	77.8
	Vejrumbro*	77	2.00×10^{-4}	0	1.429	0.23	19.5	75.3

* Sites containing datapoints from peat geological unit.

transformations that the peat soil might have experienced in its past. Moreover, especially in less decomposed peat, the pore space consists mainly of more or less intact cell structures as residuals from the former plant material. These can predominate the NMR signal but might act as dead-end pores with low to no contribution to the hydraulic water flow. These might be reasons for a general failure of the K estimation from NMR for organic soils, except for gyttja (see the next subsection).

Overall, though, it should be emphasized that a general calibration for inorganic unconsolidated geological materials is possible; this claim will be supported by justification presented later in subsection 5.5.

5.2. Is Geology-specific calibration necessary?

Our study showed that, overall, the geology-specific calibration

improved K estimation beyond the general calibration approach. Considering each specific unit, however, the results were more variable, and in some cases, the use of geology-specific calibration did not improve accuracy for K estimation. We found that the cases in which accuracy was improved had the following conditions: (1) the K and T_{2ML} ranges were not limited, e.g., that they spanned at least 1–1.5 orders of magnitude; and (2) there were enough points in the datasets to perform a statistically reliable calibration. We note that obtaining a large range of T_{2ML} values can be tricky/impossible for some geological materials such as clays that are inherently characterized by fast relaxation times. Moreover, low SNR for borehole NMR in noisy conditions, and lower relaxation time resolutions are also potential problems for these geological materials.

While reliable K estimates could not be obtained for organic peat soils, geology-specific calibration proved successful for gyttja, another organic geological material. Similar to inorganic units such as sand and clay and in contrast to peat, the application of geology-specific calibration for gyttja significantly improved K predictions relative to the general approach. Moreover, calibrating the SDR model specifically for gyttja appears necessary, as the resulting SDR coefficients differ markedly from those observed in unconsolidated inorganic materials.

Overall, we conclude that if geological data are known and available, adopting a geology-specific calibration can improve the accuracy of K estimates provided that there is a sufficient number of data points and the range of parameters is not limited. However, a geology-specific calibration for unconsolidated materials is not always necessary as long as organic units are excluded from the general calibration.

5.3. Is site-specific calibration useful?

Like the geology-specific calibration, our results showed that in some cases, site-specific calibration improved the accuracy of K estimations. The conditions under which site-specific calibration was useful mirrored that for geology-specific calibration. Calibration was improved for sites with no organic units, when the range of T_{2ML} and K values was not limited, and when a sufficient number of data points were available.

5.4. Is it necessary to include porosity in the K estimation model?

As discussed, several studies have reported that porosity has a negligible effect on K estimation in unconsolidated geological materials when using the SDR model with free-fitting b , m , and n parameters (Kendrick et al., 2021; Mashhadi et al., 2024; Maurer and Knight, 2016). Consistent with these findings, our study – across both laboratory and field datasets – frequently yielded optimal m -values of zero or very close to zero in many successful calibration cases. Even in cases where m was non-zero (i.e., porosity contributed to K estimation), T_{2ML} alone was often sufficient to compensate for the absence of porosity in the model. This suggests that porosity can be excluded without significant loss of predictive accuracy, supporting the use of the simplified SDR model as a practical and effective empirical approach for unconsolidated sediments.

Another compelling reason to exclude porosity from the SDR model comes from observations in the laboratory NMR dataset. While increasing magnetic susceptibility appears to have a weak influence on relaxation time (T_{2ML}), its impact on NMR-derived porosity is significant. In particular, porosity tends to be notably underestimated in samples with high magnetic susceptibility. This underestimation is critical, as errors in the porosity parameter directly propagate through the SDR equation and can lead to substantial inaccuracies in K estimation. Therefore, for unconsolidated sediments, excluding porosity from the SDR model not only simplifies the equation by reducing the number of variables in the equation, but also prevents some sources of error from influencing the K predictions.

5.5. General calibration for inorganic sediments: standard SDR vs. simplified SDR

To show that the effect of porosity can be compensated for by the relaxation time component in the standard SDR model (i.e., T_{2ML}), and that the use of a simplified SDR model is valid, a comparison between standard SDR and simplified SDR is necessary. Moreover, it was also hypothesized that a general calibration using inorganic unconsolidated geological materials should be successful for reliable K estimation. Hence, a bootstrapping analysis of general calibration for inorganic geological materials using standard SDR and simplified SDR models are performed for the laboratory dataset (Fig. 8; Table 4) as well as the field dataset (Fig. 9; Table 5). The color-filled histograms show the results of the standard SDR model while the black histograms illustrate the outcomes of the simplified SDR model.

The general calibration of inorganic geological units for the laboratory dataset results in a successful K estimation, as can be seen in the validation plot in Fig. 5d. The bootstrapping analysis for the standard SDR model provided an R^2 score of about 0.7, suggesting that the data fits the model well, and the K_{diff} of about 6, indicating that the K estimation error is within acceptable limits. Relatively similar values are also observed for the simplified SDR model. Fig. 8 shows that the histograms of n , R^2 score, and K_{diff} for standard SDR and simplified SDR are also quite similar. On the other hand, the distribution of b is much narrower for simplified SDR compared to standard SDR, i.e., the 95 % CIs for b ranges from 0.005 to 0.34 for the standard SDR compared to 0.0025–0.015 for the simplified SDR (more than two orders of magnitude variation compared to less than one). In addition, for the standard SDR model, the histogram of m shows a wide distribution. These facts mean that the use of the simplified SDR model results in considerably less uncertainty for the constant coefficients, with negligible effects on K estimation accuracy.

For the field dataset, the general calibration of inorganic unconsolidated geological units was also effective for K estimation, as highlighted in the validation plot in Fig. 7d. The bootstrapping analysis for the standard SDR model provided an R^2 score of 0.46 and a K_{diff} of 8.5. A few outliers, from the Norre Vosborg dataset, substantially affected the fits and excluding these points would have improved the data fit and estimation accuracy metrics. The histograms of b , n , R^2 score, and K_{diff} for the standard SDR and simplified SDR models are nearly identical (Fig. 9) as the m values were estimated to be zero during calibration of the standard SDR model. The field dataset also showed that neglecting porosity does not affect K estimation and provides a way to simplify the K estimation model.

6. Conclusion

This study investigated hydraulic conductivity (K) estimation by NMR data for unconsolidated geological materials using the SDR model. We examined 92 laboratory samples from nine sites and 198 borehole screens with borehole NMR data from eight field sites in Denmark and Germany, encompassing unconsolidated sediments (sand, clay, till) and organic materials (peat, gyttja). Based on laboratory and field datasets, the SDR model performs well for K estimation in inorganic unconsolidated geological materials. While gyttja also follows the SDR model with different constant coefficient values compared to inorganic geological materials, peat soil does not follow the SDR model. For our peat soils, relaxation time and K do not display any correlation. The reason for this might be that relaxation time variations do not reflect pore size changes alone. This aspect deserves future attention.

Our study also validated that general calibration cannot be effective for unconsolidated geological units unless it is considered exclusively for inorganic materials. In addition, it is observed that geology-specific and site-specific calibrations can significantly improve K estimations if sufficient data points are available, and T_{2ML} and K ranges are not limited. We also recommend using the “simplified SDR model” for estimating K

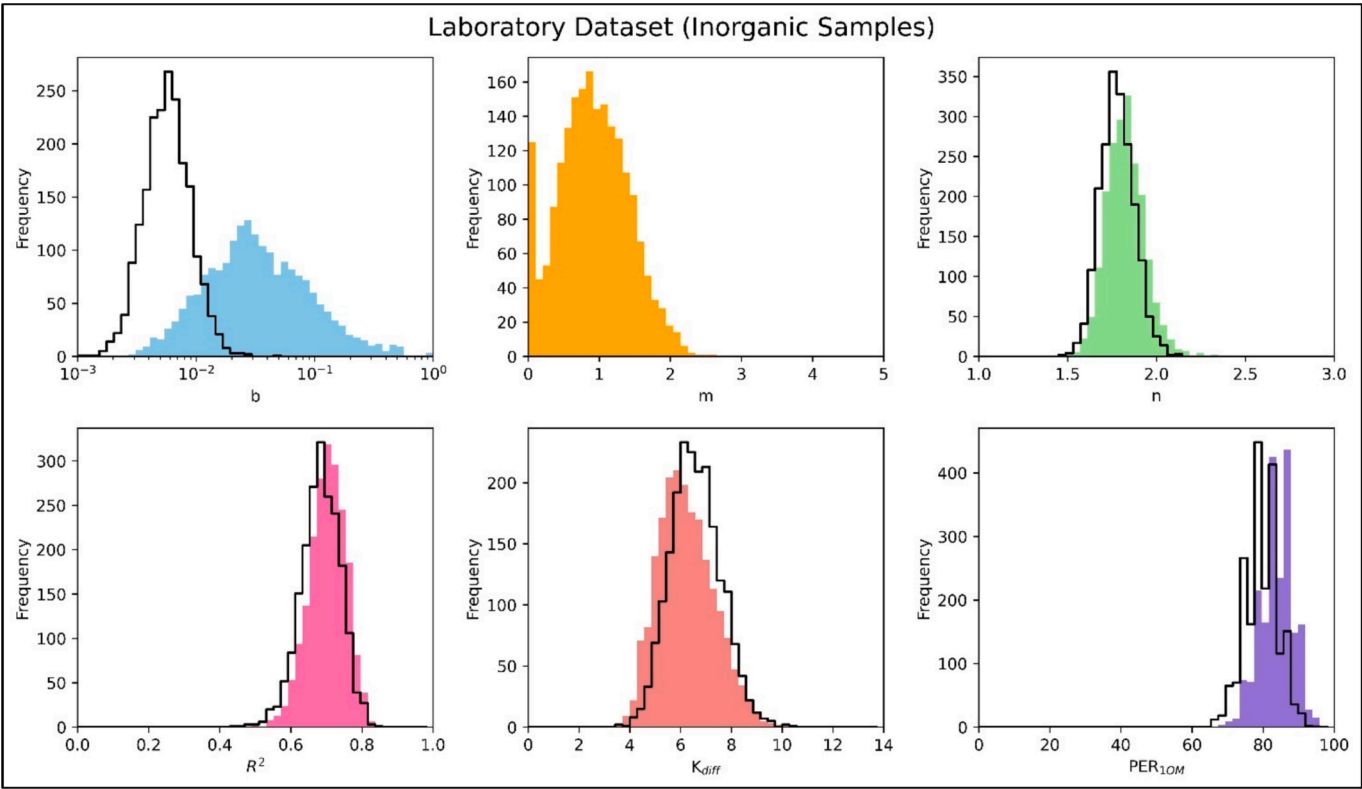


Fig. 8. Bootstrapping analysis results for the standard SDR model (color-filled histograms) and simplified SDR model (black histograms) using the inorganic geological samples. There is no black histogram for *m* since this coefficient factor is set to zero (i.e., porosity is excluded from the model).

Table 4
Statistical values (median, lower confidence level, LCL and upper confident level, UCL) derived from bootstrapping analysis of the inorganic laboratory samples (i.e., sediments) for the standard SDR and simplified SDR models.

Parameter	Standard SDR Model			Simplified SDR Model		
	Median	LCL	UCL	Median	LCL	UCL
<i>b</i>	3.0×10^{-2}	5.0×10^{-3}	3.4×10^{-1}	5.8×10^{-3}	2.5×10^{-3}	1.5×10^{-2}
<i>m</i>	0.87	0.00	1.96	—	—	—
<i>n</i>	1.81	1.63	2.05	1.78	1.61	1.99
<i>R</i> ²	0.71	0.60	0.79	0.69	0.57	0.78
<i>K</i> _{diff}	6.1	4.3	8.6	6.5	4.8	8.7
<i>PER</i> _{10M} (%)	84.9	75.3	91.8	79.5	69.9	89

in inorganic unconsolidated geological materials (i.e., $K = b(T_{2ML})^n$). First, it offers a simpler formulation to estimate *K*, involving only three parameters instead of five. Second, excluding porosity is justified by the lack of correlation between total porosity and *K*. Finally, while increasing magnetic susceptibility has only a weak influence on *T*_{2ML}, it substantially affects porosity estimation, causing significant underestimations that can directly impact *K* estimation.

CRedit authorship contribution statement

Seyyed Reza Mashhadi: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kristina Keating:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition,

Formal analysis, Conceptualization. **Rasmus Jes Petersen:** Writing – review & editing, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Andres Ospina Parra:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Stephan Costabel:** Writing – review & editing, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gulmira Beisembina:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Thomas Hiller:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Denys Grombacher:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

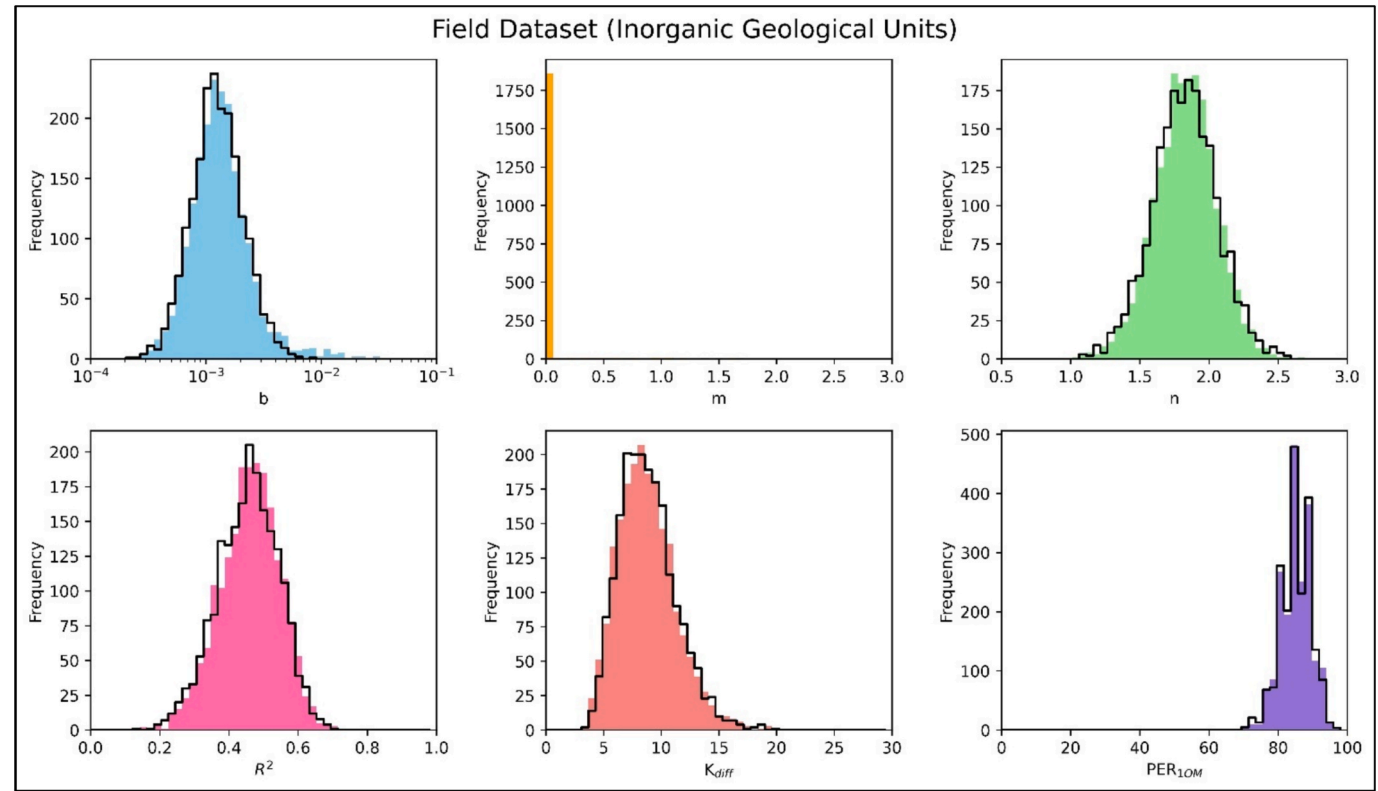


Fig. 9. Bootstrapping analysis results for the standard SDR model (color-filled histograms) and simplified SDR model (black histograms) using the inorganic geological units in the field dataset. There is no black histogram for *m* since this coefficient factor is set to zero (i.e., porosity is excluded from the model).

Table 5
Statistical values (median, lower confidence level, LCL and upper confident level, UCL) derived from bootstrapping analysis of the inorganic geological units (i.e., sediments) in the field dataset for the standard SDR and simplified SDR models.

Parameter	Standard SDR Model			Simplified SDR Model		
	Median	LCL	UCL	Median	LCL	UCL
<i>b</i>	1.3×10^{-3}	4.8×10^{-4}	6.4×10^{-3}	1.2×10^{-3}	4.6×10^{-4}	3.5×10^{-3}
<i>m</i>	0	0	1.31	—	—	—
<i>n</i>	1.83	1.37	2.27	1.83	1.35	2.28
<i>R</i> ²	0.46	0.28	0.62	0.46	0.27	0.61
<i>K</i> _{diff}	8.5	4.7	13.7	8.6	4.9	14.3
<i>PER</i> _{10M} (%)	85.3	77.3	93.3	85.3	76	92

Appendix A.: Grain size distribution data of sediment samples

Grain size distribution was also measured for sediment samples in the laboratory using a Laser Diffraction Particle Size Analyzer (LDPSA) capable of measuring 0.375 μm to 2 mm particles. Particle sizes above 2 mm were sieved out and discarded; however, only a few samples had particles larger than 2 mm. LDPSA data were used to determine the grain size distribution from the volumetric abundance of particles, and representative metrics such as *d*₁₀ are then extracted for each sample to be used for further analysis.

Hazen’s method is a common empirical approach for estimating *K* for soils and sediments based on grain size distribution (Hazen, 1892):
$$K = C.d_{10}^2 \tag{A1}$$

where *K* is permeability is in cm/s, *C* is a constant factor ranging from 1 to 1.5, and *d*₁₀ is the particle diameter in millimeters such that 10 % of all sample particles are finer by weight (Zaroei and Fereidooni, 2023). This empirical equation was developed empirically using washed sediment samples with grain sizes > 10 μm (Hazen, 1892), and it performs well for granular materials like sands and gravels (Carrier, 2003). However, it becomes less accurate and unreliable for fine-grained soils (e.g., clays), where permeability is influenced by particle shape, cohesion and electrochemical interactions between clay minerals (Mitchell and Soga, 2005). Other models have been developed based on different metrics extracted from grain size distribution to estimate *K* (Zaroei and Fereidooni, 2023), but the most common is Hazen’s method and so this is the approach used here.

After measuring grain size distribution by the LDPSA method, the volumetric-based distribution of grains was generated for each sediment sample

and standard representative metrics (e.g., mean, d_{10} , etc.) are extracted from the distribution. An example of grain size distribution data from a sand sample from the Norre Vosborg site is presented in Fig. A1, which highlights a unimodal distribution with a mean grain size of 316 μm , and an effective diameter (d_{10}) of 162 μm . The sand, silt, and clay fractions are determined based on the threshold values suggested by Peng et al. (2021).

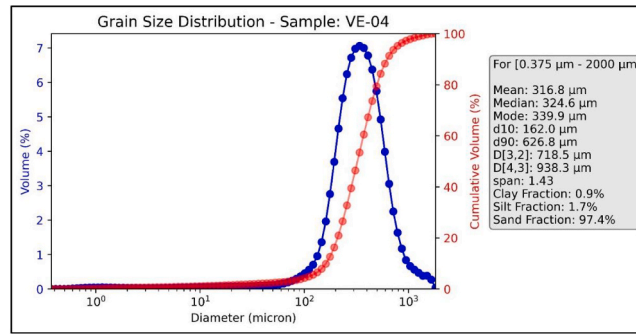


Fig. A1. Grain size analysis results of a sandy sample from Norre Vosborg.

In the calibration of Hazen's equation, one anomalous sample was recognized which exhibited a significant mismatch between the estimated and measured K values so due to its obvious outlier behavior, it was excluded from the analysis in this section. According to curve fitting (Fig. A2), the C is estimated to be 1.00, which is in the expected range of 1 to 1.5 (Zaroei and Fereidooni, 2023). The R^2 score is 0.74, and the K estimation error from grain size distribution data is highlighted by the K_{diff} of 5.2. Moreover, it seems that the samples with high K values (i.e., $K > 1 \times 10^{-5} \text{ m/s}$) are mostly overestimated by Hazen's model while samples with small K values (i.e., $K < 2 \times 10^{-6} \text{ m/s}$) are often underestimated by the model. A possible reason is that our samples include sediments with fine particles (e.g., clay minerals) while Hazen's model is developed for silts, sands, and gravels, i.e., particle sizes larger than 10 μm (Hazen, 1892).

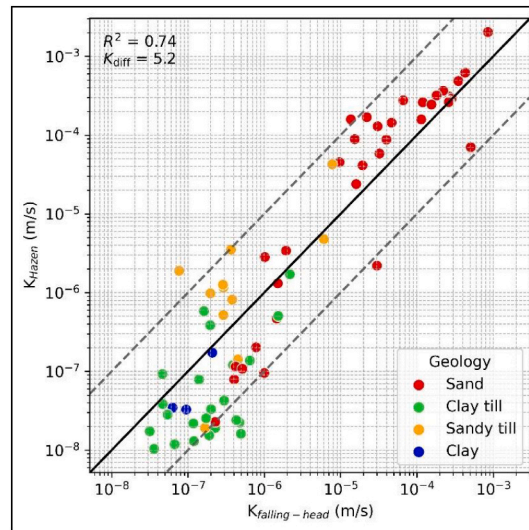


Fig. A2. Estimated hydraulic conductivity by Hazen's equation vs. measured hydraulic conductivity in the falling head experiments.

The result of bootstrapping analysis for Hazen's model resulted in the median value of C , R^2 and K_{diff} are 1.30, 0.75, and 4.9, respectively (Fig. A3, and Table A1). The 95 % CIs for the parameter C span a pretty narrow range from 1 to 1.82. The K estimation by Hazen's method for our set of sediment samples showed that the method works pretty well, but the estimation accuracy might be affected when dealing with clay-rich mixtures. Also note that LDPSA provides volume-based distribution while the defined d_{10} in the Hazen's method was derived from a weight-based distribution (Hazen, 1892), meaning that inaccuracies may occur if density distribution is significantly variable across grain sizes in the sample.

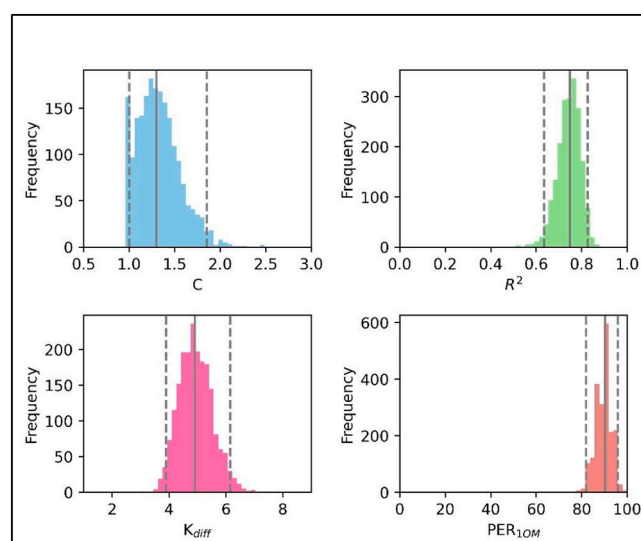


Fig. A3. Bootstrapping analysis results for Hazen's model.

Table A1

Median and 95 % CIs derived from bootstrapping analysis of the grain size distribution data of sediment samples for Hazen's model.

Parameter	Median	LCL	UCL
C	1.30	1.00	1.82
R^2	0.75	0.63	0.83
K_{diff}	4.9	3.8	6.2
PER_{10M}	90.3	81.9	95.8

Data availability

The data can be downloaded from this link: <https://doi.org/10.5281/zenodo.15871335>.

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